

BORDER OPTICS — POORI RESEARCH SAMJHO

PDF 21 of 22 — Research Paper, Part 2

**Results, Discussion, Limitations, Future Work, Conclusion
(Section 4 se 8 tak)**

Sakshi ke liye, Sakshi ki apni research par. PDF 20 mein humne is paper ka poora 'setup' seekha — sawaal, theory, tareeka. Ab isi PDF mein wo dekhte hain jo us tareeke ko apply karne se mila — Results. Ye paper ka wo hissa hai jahan sabse zyada imaandaari dikhani padti hai, kyunki yahin numbers khud bolte hain, chahe wo convenient hon ya na hon. Ye is poori research ka sabse dense hissa bhi hai — isliye ye PDF baaki PDFs se lamba hoga, kyunki 'poora samajh aana' hamesha page-count se zyada zaroori raha hai.

Scope: BO_Research_Paper.md ki Section 4 (Results, 4.1 se 4.10), Section 5 (Discussion), Section 6 (Limitations, 6.1 se 6.12), Section 7 (Future Work, 7.1 se 7.7), Section 8 (Conclusion), aur References ka ek chhota overview.

Section 4 — Results: Wahi Tareeka, Ab Jo Mila

4.1 — Village Coverage Aur Sample Composition

Geocoding pipeline se 559 mein se 258 villages locate hue. Arunachal Pradesh mein bahut kam civilian entries mili. Isse aage check karne ka koi tareeka bhi nahi hai, kyunki border par inhabited points ki sirf concrete lists BRTF camps, army staging huts, aur labour camps se aati hain — inme se koi bhi kisi civilian geospatial database mein hai hi nahi. Ye data mein koi bada gap nahi hai, dusre shabdon mein. Ye ek seedhi tasveer hai ki ek securitized border par 'village-level development' asal mein dikhta kaisa hai.

4.2 — Built-Up Area Change: Dono Windows Mein Ab Koi Significant Change Nahi, Poore Data Ke Saath

Do compositing windows, ab dono ka ek hi conclusion, is test ke asli sawaal par: Full-year — 251-village core sample mein koi significant *increase* nahi (Wilcoxon signed-rank, one-sided, alternative='greater', $p = 1.000$), aur ye number kabhi nahi hila, is study ke kisi bhi version mein. Summer-matched (June-September) — bhi koi significant increase nahi, ab jab sample poora hai ($n = 251$, $p = 0.435831$).

Ek Naya, Zaroori Detail — ' $p = 1.000$ ' Ka Asli Matlab

EXAMPLE — One-Sided Vs Two-Sided Test

Socho tum ek race judge ho, aur tumhara kaam sirf ye check karna hai ki 'kya Runner A, Runner B se AAGE nikla' — agar Runner A peeche bhi reh jaaye, tumhara jawab sirf 'nahi, aage nahi nikla' hoga, chahe wo kitna bhi peeche kyun na reh gaya ho. Yahi ek 'one-sided' test hai — H_1 sirf ye poochta hai ki kya VVP-I ne development *badhaya*, kisi bhi disha mein 'change' nahi. Isliye iska design kabhi bhi ek decrease ko positive finding ki tarah flag karne ke liye bana hi nahi tha. Lekin agar tum wahi race dono directions mein judge karo — 'kaun aage hai, chahe A ho ya B' — to ye ek 'two-sided' test ban jaata hai, jo dono directions ko barabar toltta hai.

Yahi wajah hai ki paper 'no significant change' ko bina qualify kiye chhod nahi deta. Full-year window ka median change -0.017 hai, zero ke paas nahi, aur 251 core villages mein se 75.7% ek decrease dikhte hain, increase nahi. Wahi data agar two-sided test se chalaya jaaye, to $p = 1.4 \times 10^{-15}$ milta hai — ek highly significant *decrease*. Ye ek asli, measured pattern hai data mein, koi rounding-artifact nahi, aur ise seedha kehna zaroori hai, na ki 'no significant change' ya 'barely negative at the median' jaisi bhasha mein chhupana, jo is section ke pehle versions use karte the aur is distribution ki asli ek-tarfa-pan ko kam karke dikhte the.

Ye H_1 ke jawab ko nahi badalta: is study ne poocha tha ki kya VVP-I villages measurably zyada built-up hue, aur jawab hai nahi, kisi bhi window mein. Lekin

full-year window ek *flat* built-environment bhi nahi dikhata. Ye ek aisa environment dikhata hai jo, is measure par, zyadatar villages ke liye peeche gaya, kisi ke liye aage jaane se zyada, usi period mein. Ye decrease kya reflect karta hai (satellite-visible seasonal ya archive effects jinka zameen ki asli condition se lena-dena nahi, ek genuine lekin VVP-I se alag regional pattern, ya kuch aur) — ye is study ka design distinguish nahi kar sakta, aur yahan koi koshish nahi ki gayi hai ise explain-away karne ki.

Test	Full-Year Window	Summer-Matched Window
H1 one-sided (increase?)	p = 1.000	p = 0.435831, n = 251
H1 two-sided (change?)	p = 1.4×10^{-15} (decrease)	—
Median change	-0.017	—
% villages decreasing	75.7%	—

History Woven In — Wahi 154-Village Wala Reversal, Ab Poori Tarah

Summer window ka number hi wo hai jo badla. Kyun badla, ye maayne rakhta hai, isliye seedha bataya ja raha hai, kuch chhupaya nahi ja raha. Is test ka ek pehla version, Development Log Entry 22 se pehle chalaya gaya, $p < 0.000001$ report karta tha. Lekin wo sirf ek 154-village subsample par tha, kyunki us waqt Sikkim ke geocoded sabhi 31 villages summer window ke liye zero cloud-free Sentinel-2 imagery ke saath wapas aaye the, aur poori tarah bahar gir gaye the. Wo 'zero' data ke baare mein koi fixed fact nahi tha. Wo bas Sentinel-2 archive kitna backfill ho chuka tha uska ek snapshot tha, us ek din jab extraction chala. Ek baad ki, complete re-extraction — same collection, same cloud mask, same 500m buffer, same date ranges, control-group ke summer-data ke saath usi din pull ki gayi — ne is baar sabhi 31 Sikkim villages ke liye non-null NDBI values di. Unka mean summer NDBI change (-0.0347) teeno core states mein sabse negative hai (Figure 2), aur yahi pooled result ko highly-significant se ek saaf null tak kheechta hai. Poori detail, plus do extraction-script bugs jo iske process mein mile aur fix hue, Development Log Entries 21 aur 22 mein hain. 154-village wala result, aur is paper ke pehle drafts mein uspar jo bhi bana tha, sirf thoda refine nahi hua — poori tarah is section ke current numbers se replace ho gaya hai.

4.3 — Night-Lights Change: Ek Stable Null

Night-lights ek steady kahani sunate hain is do-point comparison mein: dono windows mein koi significant change nahi (full-year $p = 0.050$, seedha threshold ke kinare par; summer-matched $p = 0.9999$, ek saaf null). Snow aur vegetation is proxy ko waise disturb nahi karte jaise NDBI ko kar sakte hain, aur yahi ek wajah hai ki night-lights ko

dono proxies mein zyada bharosemand maana jaata hai — NDBI utna hi achha hai jitne bands use feed karte hain. Full-year ka borderline case ($p = 0.050$) ek log1p-transformed version par bhi check hua (kyunki VIIRS radiance right-skewed hota hai) — wahan bhi non-significant hi raha ($p = 0.068$), isliye log-transform bhi kuch nahi palatta. 'Steady' sirf is do-point comparison ke liye sahi hai, underlying year-by-year path ke liye nahi, jo Section 4.7 dikhata hai asal mein flat nahi hai.

4.4 — Budget Vs Outcome: Ek Teen-State Descriptive Comparison (RQ2)

State	Mean NDBI Change (Summer)	Sanctioned Budget	Projects
Arunachal Pradesh	+0.0023	Rs. 2,749.74 crore	2,082
Uttarakhand	+0.0125	Rs. 270.58 crore	200
Sikkim	-0.0347	Rs. 188.90 crore	63

Teeno core states ke paas ab valid summer-window NDBI coverage hai (Section 4.2 — ye pehle drafts mein sirf do-state comparison thi, Sikkim ka data wapas aane se pehle). Summer window ke across, Spearman correlation: $\rho = 0.500$, $p = 0.667$ — $n = 3$ par kisi bhi powered test ke paas-paas bhi nahi, isliye ye hamesha descriptive-only rehta hai.

Arunachal Pradesh ka budget Uttarakhand se lagbhag das-guna hai, aur wo ek chhote measured-change se milta hai. Sikkim, teeno mein sabse chhote budget wala, ek negative change se milta hai. Ye H2 ki prediction se fit baithta hai ki outcome budget ke saath scale nahi karta. Lekin yahi comparison full-year window ke teeno states par chhalao (Sikkim wahan kabhi missing nahi thi), to $\rho = -0.500$ milta hai — bilkul ulta sign. Sirf teen data-points ke saath, aisi sign-flip iska saboot nahi ki relationship ulti ho gayi. Ye saboot hai ki teen states yahan kuch bhi kehne ke liye kaafi hi nahi hain. Dono directions report ki gayi hain. Jo zyada convenient hoti usko chunna aasan hota — wo nahi hua.

4.5 — Border Proximity Testing (H3)

Full-year window mein, built-up change border/LAC se distance ke saath bilkul correlate nahi karta ($\rho = 0.049$, $p = 0.443$) — wahi null jo is paper ke har pehle version mein mila. Lekin summer-matched window mein, complete 251-village sample par, ab karta hai: $\rho = 0.291$, $p = 0.0000026$. Border se door wale villages, border ke paas wale villages ke muqable, bada NDBI increase (ya chhota decrease) dikhate hain — jo H3 ki prediction se ulti disha hai; securitization-driven prioritization ko paas wale villages ko aage rakhna chahiye tha, peeche nahi. Ye ek bilkul naya result hai. Pehle wale,

154-village summer sample par, yahi test ek saaf null wapas deta tha ($\rho = 0.038$, $p = 0.641$), aur is section ke aur Section 4.9 ke ek pehle version ne ise 'ek sachmuch stable null... kisi bhi window mein' kaha tha. Ye description ab sahi nahi baithti, Sikkim ka data shaamil hone ke baad (Section 4.2, Development Log Entry 22).

Section 4.2 ke summer NDBI change result jaisa hi, ye bhi ab leave-one-district-out aur randomization-inference checks se guzar chuka hai, aur dono mein zinda bachta hai (Section 7.6, Development Log Entry 32). Iske p-value ka ek spatially-corrected version bhi ab maujood hai (Section 6.11), naive number se kaafi kam extreme, lekin phir bhi significant.

Night-lights ki border-proximity is data-refresh se chhoo bhi nahi hui. Full-year window mein abhi bhi significant hai ($\rho = -0.252$, $p < 0.0001$, paas wale villages zyada change dikhate hain, H3 se match karta hai), summer-matched window mein abhi bhi null hai ($\rho = -0.071$, $p = 0.266$).

History Woven In — UTM Zone Se Geodesic Distance Tak

Is section ka ek pehla version in correlations ko nikaalne ke liye har village ko ek hi UTM zone (44N) mein project karta tha, jo sirf apne central meridian (81°E) ke paas accurate hota hai. Is study ke villages roughly 77°E se 97°E tak phaila hain, isliye wo tareeka un villages ke liye distance overstate kar raha tha jo 81°E se door the — Arunachal Pradesh ke sabse purvi villages ke liye lagbhag 3% tak, Uttarakhand ke liye lagbhag kuch bhi nahi. Distance ab geodesically compute hoti hai (WGS84 ellipsoid), koi zone-dependency nahi (Development Log Entry 18). Ye fix har ρ aur p ko thoda hilata hai, lekin koi conclusion nahi badalta: kuch bhi pehle significant nahi tha aur ab bhi nahi hai, sivaay full-year lights correlation ke, jo pehle bhi aur baad bhi significant tha.

4.6 — Control Group Difference-in-Differences: Kisi Bhi Window Mein Koi Gap Nahi Tiktha (H4)

History Woven In — Control-List Ki Poori Kahani, 753 Se 721 Tak

Is section ka ek zaroori note: control-group list mein do pehle ke defects mil chuke hain, fix ho chuke hain, aur fix live data ke against chalaya jaa chuka hai (Development Log Entries 23 se 25). Ek pehla draft in DiD estimates ko ek 753-village control-list ke against report karta tha, jisme ek cross-district-matching bug se 18 duplicate coordinate rows the (fix hoke, Entry 22 ki 735-village list bani). Us 735-village list mein bhi 20 aise villages nikle jo treated villages ke hi exact coordinate-duplicates the, ek alag naam-spelling ya script mein, plus 3 aur jo ek official lekin ungeocoded VVP-I priority-village se naam-match karte the. Matlab control-group, kuch hisse mein, khud treated villages ko apne khilaaf gin raha tha (Entry 23). Dono problems code mein fix hue — ek geodesic 50m coordinate-proximity exclusion plus ek poori official-naam-list

exclusion (Entry 24) — aur phir wo fix hamesha ke liye carry-out hua (Entry 25): control-list ab 732 villages par khadi thi, independently zero exact-coordinate-overlap aur zero official-naam-match par check ki gayi.

Ek aur correction (Development Log Entry 37): Section 6.10 ki limitations-table mein flag hui district-verification gap, do (teen mein se) affected districts ke liye band hui. `select_control_villages.py` live Nominatim/Overpass access ke saath dobara chalaya gaya, is baar Tawang aur Pithoragarh ke liye asli boundary-polygon resolve hui (pehle dono sirf bbox-fallback match the), aur 11 Tawang candidates hataye gaye jo polygon-check ne confirm kiya ki kabhi us district ke andar the hi nahi. Control-list ab 721 villages par khadi hai, aur is paper ka har DiD, baseline-balance, triangulation, aur wild-cluster-bootstrap number isi ke against dobara nikala gaya hai.

Ab Numbers, 721-Village Corrected List Par

Outcome / Window	DiD Coefficient (95% CI)	Cluster-Robust p	HC3 p
Summer NDBI	-0.00275 [-0.0114, +0.0059]	0.53098	0.69943
Full-Year NDBI	+0.00668 [-0.0057, +0.0191]	0.29102	0.61263
Summer Lights	+0.01798 [-0.0583, +0.0943]	0.64393	0.70657
Full-Year Lights	+0.03460 [-0.0558, +0.1250]	0.45334	0.70730

Summer NDBI gap ek reversal hai original, contaminated aur incomplete extraction se, jo ek significant positive gap dikhati thi. Full-year lights ka gap sabse zyada maayne rakhta hai: ye is study ka ek-hi-time surviving significant control-group result tha, jo ab null ho gaya hai (down from pre-Entry-25 estimate ka +0.1653, $p = 0.0364$). Wo pehla, significant-dikhne-wala estimate us contaminated control-pool par partly tika hua tha jise Entry 23 ne pakda aur Entry 24 ne code mein fix kiya: 20 control-villages jo physically wahi settlements the jo treated villages, plus 3 aur jo officially VVP-I villages ki tarah unke listed naam se list mein the. Jo 'surrounding non-priority region ke night-lights steady rehte hue treated villages ke declining' jaisa dikh raha tha, uska ek hissa asal mein, un 23 rows ke liye, treated-group ki apni trajectory ko usi regression ke control-side par doosri baar bhi gin liya jaana tha. Ye contamination hamesha ke liye hataao (Development Log Entry 25), aur gap bas gayab ho jaata hai. Yahi wo exact artifact hai jiska Entry 23 ne pehle hi khadka diya tha, aur fix karke, result ko gayab hote dekh kar confirm hua. Ye kabhi ek asli lekin fragile effect nahi tha jo chupke se fade ho raha tha.

Baseline Imbalance — Resolve Hua, Phir Dobara Khula

Baseline-imbalance ki tasveer Development Log Entry 37 ke district-verification-fix ke saath dobara badli — aur 'zyada resolved' ki disha mein nahi. Full-year: NDBI significantly imbalanced nahi hai (treated mean -0.18459 vs. control -0.18327, $n=251/721$, $p = 0.18118$), lekin night-lights — jo Entry 25 ke contamination-fix se pehle is study ka sabse bada imbalance tha, aur jise Entry 25 ne non-significant tak la diya tha ($p = 0.08976$) — corrected 721-village list par dobara imbalanced hai: treated mean 0.42821 vs. control 0.43493, $n=251/721$, $p = 0.03558$. Summer: NDBI ($p = 0.46099$) aur night-lights ($p = 0.08767$) balanced hi rehte hain, 732-village version se lagbhag same. Chaar mein se teen baseline-balance checks treated aur control villages ke beech statistically indistinguishable hain 2021 mein; chautha, full-year night-lights, nahi hai. Section 6.7 mein hum poori tarah dekhenge is ka is design ke parallel-trends-assumption ke liye kya matlab hai.

Leave-One-Out, Randomization Inference, Aur Log1p Check

Do aur checks probe karte hain ki kya ek DiD estimate sirf 14 district-clusters ke saath bharosemand hai, chahe wo estimate khud significant ho ya na ho. Ab kuch bhi significant bacha nahi hai stress-test karne ke liye, dono checks bas null ko confirm karte hain. Summer window: regression ko 14 baar rerun karna, har baar ek district hata kar (extended_robustness_checks.py), NDBI ko 0 mein se 14 reruns mein significant chhodta hai (coefficient range -0.00564 se +0.00046), aur night-lights ko 0 mein se 14 mein (range -0.00949 se +0.04238). Randomization inference (2,000 permutations, treatment ko district ke andar reshuffle karke) deta hai $p = 0.22939$ NDBI ke liye, $p = 0.49325$ night-lights ke liye. Full-year window: NDBI kabhi significant tha hi nahi, isliye alag se re-tabulate nahi hua, lekin full-year *night-lights* result — jo yahan maayne rakhta tha kyunki wo significance claim karta tha — wo bhi ab saaf-saaf fail hota hai: leave-one-district-out ise 0 mein se 14 significant reruns tak le jaata hai (range +0.00066 se +0.05818, pehle contaminated data par 10 mein se 14 significant, range +0.1161 se +0.2166 ke against), aur randomization inference observed +0.03460 coefficient ko $p = 0.33533$ par rakhta hai, ek permutation-mean +0.00019 (SD 0.03516) ke against — bilkul apni null-distribution ke andar, tail se kaafi door.

Ek caveat, har pehle version jaisa hi: VVP-I priority-village status haqiqat mein randomly assign nahi hui thi, programme ne ye villages administratively chuni. Ye procedure sirf test karti hai ki observed treated-vs-control gap, district ke andar label-exchangeability ke ek null ke against, ajeeb dikhta hai ya nahi — ye check karta hai ki significance-test sirf 14 clusters ke saath bharosemand hai ya nahi. Ye random-assignment ka substitute nahi hai.

Kyunki VIIRS night-lights radiance right-skewed hota hai, night-lights DiD ko log1p-transformed radiance par bhi rerun kiya gaya, dono windows. Summer log-scale par bhi null rehta hai (raw $p = 0.64393$; log1p $p = 0.72112$). Full-year result pehle dono

scales par significant tha (raw $p = 0.0364$; $\log_{10} p = 0.0374$), aur usi baat ko saboot ki tarah dikhaya jaata tha ki result sirf ek right-skew-artifact nahi hai. Ab dono scales par null hai: raw $p = 0.45334$; $\log_{10} p = 0.61112$. Ye raw result se match karta hai, contradict nahi karta, aur pre-Entry-25 result jis last angle se solid dikhta tha, usko bhi band kar deta hai.

4.7 — Multi-Year Trend: Ek Non-Monotonic Recovery Pattern

Ek teesra saal add karo, aur '2021 vs 2025 ki kahani' zyada complex ho jaati hai. Summer-matched window mein, mean NDBI 2021 se 2023 tak girta hai (mean change = -0.0231 , Wilcoxon $p = 1.000$, ek decline, significant increase ki jagah), phir 2023 se 2025 tak tezi se badhta hai (mean change = $+0.0222$, $p < 0.000001$). Phir bhi, poora 3-saal ka linear trend significant nahi hai (mean per-village slope = $-0.0002/\text{year}$; Wilcoxon $p = 0.442$; village-fixed-effects panel regression $p = 0.728$). Full-year window bhi wahi shape dikhati hai — decline, phir 2023-2025 mein ek significant increase ($p = 0.000001$) — lekin yahan dono statistical methods overall trend par asehat hain: Wilcoxon $p = 1.000$, panel regression $p = 0.000001$. Ye farq is wajah se hai ki panel regression 2023-2025 ki recovery ko earlier decline se zyada weight deta hai, jo Wilcoxon bilkul nahi karta. Ye Section 4.2 ke headline number ko padhne ke liye maayne rakhta hai: wahan cite hua asli 2021-vs-2025 result asal mein ek 2023-2025 recovery hai jo ek do-point comparison mein dikh rahi hai, aur earlier downturn ko padhte waqt bhoolna nahi chahiye.

Night-lights, jise Section 4.3 'ek steady kahani' kehta hai us do-point 2021-vs-2025 comparison par, neech se actually steady nahi hai, jab wahi teesra saal add hota hai — aur ye seedha kehna zaroori hai, Figure 8 ki apni night-lights series ko bina discuss kiye chhodne ke bajaye. Full-year window mein, lights significantly badhte hain 2021 se 2023 (mean change $+0.0394$, Wilcoxon $p = 2.2 \times 10^{-6}$) aur phir badhte rehte hain, lekin ab significantly nahi, 2023 se 2025 (mean change $+0.0466$, $p = 0.748$) — overall 2021-2025 do-point comparison jo Section 4.2 report karta hai ($p = 0.050$) threshold ke bilkul kinare par baithta hai, kaafi hadd tak isliye kyunki zyaadatar multi-year movement 2023 tak ho chuka tha. Summer window mein, pattern timing mein ulta hai: lights 2021 se 2023 significantly move nahi karte (mean change -0.0665 , $p \approx 1.000$) lekin phir 2023 se 2025 tezi se aur significantly badhte hain (mean change $+0.0837$, $p = 1.7 \times 10^{-11}$) — wahi 'recovery drives the two-point number' shape jo Figure 8 ki NDBI series dikhati hai, bas ek alag outcome aur window ke liye.

4.8 — Buffer Radius Sensitivity: Teeno Radii Par Null, Same-Day Extraction Confirmed

Buffer Radius	Mean NDBI Change	Wilcoxon p (NDBI)	Wilcoxon p (Lights)
250m	+0.00175	0.1214	0.99998

500m	-0.00090	0.4358	0.99999
1km	-0.00293	0.8976	1.00000

Teeno buffer-radii sehmata hain, aur ye sehmata ek null hai. Ek pehla version is section ka ek significant 500m result report karta tha ($n=154$, $p < 0.000001$) jo 250m ya 1km par nahi tikta tha — lekin wo 500m figure wahi incomplete, pre-Entry-22 summer-extraction thi jo Section 4.2 pehle hi cover karta hai, isliye wo bhi usi ke saath supersede ho jaata hai.

Is section ke ek pehle version mein ek archive-timing gap bhi flag hua tha — 250m aur 1km extractions 500m primary-run se ek alag din par pull hue the. Wo ab band ho gaya hai: teeno radii ek hi din back-to-back re-extract hue (Development Log Entry 31), aur numbers mushkil se hile (250m: $p = 0.126$ se 0.1214 ; 500m aur 1km lagbhag unchanged 0.436 aur 0.899 se 0.8976). Archive-timing confound principle mein sach nikla. Ye bas practice mein is comparison ke liye maayne nahi rakhta, poori kahani Section 6.9 mein hai.

Isi same-day re-pull mein ek aur cheez dikhi jo pehli, timing-inconsistent comparison ne miss kar di thi: NDBI ka mean change radii ke across consistently signed bhi nahi hai — 250m par positive, 500m aur 1km dono par negative, chahe teeno mein se koi bhi significant ke aas-paas bhi nahi hai. Ye is section ke null conclusion ko fit karta hai, use complicate nahi karta — ek solid effect, jo maayne rakhne layak bada ho, kisi bhi buffer-width mein same disha dikhata, aur ye bas nahi dikhata.

4.9 — Robustness Summary

Sabhi chaar core tests ko dono windows ke against line-up karo, ab complete data par, aur ek saaf pattern dikhta hai: H1 (built-up change) dono windows mein ek null hai (full-year $p = 1.000$; summer-matched $p = 0.436$). Ye pehle wo ek test tha jo windows ke beech palat jaata tha, aur ab wo ek hai jo dono mein sehmata hai. H3 night-lights proximity abhi bhi $p = 0.05$ ki line ko windows ke beech cross karta hai, har pehle version jaisa hi — significant full-year ($p < 0.0001$), null summer-matched ($p = 0.266$). Jo badla hai wo H3 NDBI proximity hai: pehle ye dono windows mein ek stable null tha, aur ab ye usi line ko ulti disha mein cross karta hai — null full-year ($p = 0.443$), significant summer-matched ($p = 0.0000026$).

Sections 4.6 se 4.8 ke checks ko H1 aur control-group DiD ke upar stack karo, aur tasveer ab uniformly negative hai us summer-window NDBI-wali kahani ke liye jo is paper ke pehle versions ne build ki thi: ye control-group comparison mein fail hoti hai ($p = 0.53098$, tha $p = 0.00033$), leave-one-district-out mein fail hoti hai (0/14 significant, tha 14/14), randomization-inference mein fail hoti hai ($p = 0.22939$, tha $p = 0.0005$), aur buffer-radius sweep mein bhi fail hoti hai, ab teeno radii same-day-pulled ke saath

confirm ($p = 0.1214/0.4358/0.8976$ 250/500/1000m par, pehle teeno par significant thi). Full-year night-lights gap — wo ek control-group DiD result jo Entry 22 ke baad se kisi na kisi specification ke under significant raha tha — wo bhi ab null hai ($p = 0.45334$), Entry 23 ke pakde control-group-contamination ko live data ke against theek se fix karne ke baad (Entry 25), aur district-verification-fix (Entry 37) ke baad bhi null rehta hai. Ise ek documented-lekin-uncorrected-defect ki tarah baitha nahi chhoda gaya, ise fix kiya gaya: koi bhi control-group DiD result, kisi bhi outcome ya window ke liye, ab kisi bhi specification ke under significance clear nahi karta.

Naya summer H3 NDBI proximity result ($\rho = 0.291$, $p = 0.0000026$) ab leave-one-district-out aur randomization-inference checks se guzar chuka hai, aur dono mein zinda bachta hai (Section 7.6, Development Log Entry 32). Iski spatially-corrected significance Section 6.11 mein hai.

Holm-Bonferroni Correction — 8 Headline Tests Ke Across

H1 aur H3 tests ke aath headline p-values ($4 \text{ tests} \times 2 \text{ windows}$) ko ek-ek karke multiple-testing ke liye correct nahi kiya jaata, kyunki wo alag-alag research-questions ka jawab dete hain — lekin ek paper-wide Holm-Bonferroni correction (family-wise $\alpha = 0.05$) in aathon ke across chalta hai (holm_correction.py, method='holm'). Complete data par, do results ise survive karte hain, aur wo is paper ke pehle versions jaise do nahi hain. Full-year night-lights proximity result survive karta hai, jaisa ye hamesha se karta aaya hai (raw $p = 5.48 \times 10^{-5}$, Holm-adjusted $p = 3.83 \times 10^{-4}$). Summer window NDBI proximity result is list mein naya hai (raw $p = 2.64 \times 10^{-6}$, Holm-adjusted $p = 2.12 \times 10^{-5}$), aur correction ye fact kam nahi karta ki ye ek fresh, un-stressed finding hai — ye bas confirm karta hai ki ye ek-saath-aath-tests-chalane ka artifact nahi hai. Jo result pehle is correction ko survive karta tha, summer-window H1 NDBI-change test, wo ab raw $p = 0.05$ threshold bhi clear nahi karta ($p = 0.436$), isliye multiple-testing ke liye use correct karne ka sawaal hi nahi uthta.

Ye saaf karna zaroori hai ki DiD family ko alag se kyun judge kiya jaata hai, us same aath-test-correction mein fold karne ke bajaye: H1/H3 correlational hain aur har ek treated-only, ek test per outcome per window, jawab dete hue 'kya ye bilkul bhi badla.' H4 ke DiD estimates already district-clustered standard-errors carry karte hain, within-district-correlation account karte hue, aur iske upar, ek exact wild-cluster-bootstrap (Section 4.6) jo wahi asymptotics assume nahi karta jo ek simple Holm-correction karta. Ek family-wise correction ko un tests ke across combine karna jinpe pehle se do alag robustness-treatments lagi hain, wahi false-positive-defense ko do incompatible tareekon mein double-count karega, ek genuinely alag defense add karne ke bajaye. Phir bhi, completeness ke liye, Holm-Bonferroni seedha is study ke chaar primary DiD p-values par apply kiya gaya — sab kuch 1.000 tak adjust ho jaata hai, wahi conclusion jo wild-cluster-bootstrap aur har test ka apna p-value pehle hi de chuka tha.

4.10 — Independent Triangulation: SAR Aur Dynamic World Cross-Checks

Section 7.1 ka ek pehla version ek Sentinel-1 SAR cross-check ko future-work ki tarah list karta tha — external review ne isse ek aisa tareeka bataya tha jisse primary NDBI result ko ek sachmuch alag sensor ke against test kiya ja sake, sirf usi imagery par ek alag formula chalane se zyada. Wo check ab chal chuka hai, plus ek doosra, unplanned check: Google Dynamic World ka 'built' probability band, ek independently-trained land-cover classifier (Development Log Entry 26 mein poori extraction aur verification ka record hai, jisme ek incomplete-extraction-bug bhi shaamil hai jo mila aur fix hua — ek stalled checkpoint jisne control-group ki summer-window SAR-file ko sirf 732 mein se 50 villages tak complete chhoda tha, kisi bhi result report hone se pehle pakda gaya, aur completion tak re-extract kiya gaya).

SAR — Ye Poori Study Ka Sabse Mazboot Confirmation

Sentinel-1 VV aur VH backscatter (dB), usi 500m buffer aur before/after windows par extract hua jaisa is study ka har doosra test — har specification par null wapas aata hai: full-year VV (treated-only mean change -0.09879, $n=251$, Wilcoxon $p=1.00000$; control-group DiD coefficient +0.00551, 95% CI [-0.0437, +0.0547], $p=0.82614$), full-year VH (mean change -0.09873, $p=1.00000$; DiD -0.00402, 95% CI [-0.0619, +0.0539], $p=0.89164$), summer VV (mean change -0.17734, $p=1.00000$; DiD +0.01897, 95% CI [-0.0372, +0.0751], $p=0.50768$), summer VH (mean change -0.15353, $p=1.00000$; DiD +0.00012, 95% CI [-0.0791, +0.0793], $p=0.99772$). Is poore section ka, ye Sections 4.2 aur 4.6 ke conclusions ke liye sabse mazboot confirmation hai, kyunki SAR koi alag formula nahi hai jo NDBI wale hi pixels par chalta hai — ye ek bilkul alag satellite-constellation hai (C-band radar, optical nahi), us cloud-cover aur snow-contamination se bilkul untouched jinhone is study ke compositing-window-design ko shuru mein zaroori banaya (Section 2.4) aur jinhone is study ke do shuruaati extraction-bugs ki wajah bani (Development Log Entries 21 se 22). Ek alag, independent sensor ko usi 251 treated aur 721 control villages par point kiya gaya, usi do windows mein, aur usne wahi kuch-nahi paaya jo NDBI ne paaya.

Dynamic World — Ek Alag Kahani

Dynamic World ka 'built' probability ek alag kahani sunata hai, lekin. Dono windows treated-villages mein control ke mugable ek chhota lekin statistically-significant increase dikhate hain: full-year (treated-only mean change +0.00673, ek 0-se-1 probability scale par, $n=249$, Wilcoxon $p<0.000001$; control-group DiD coefficient +0.00262, 95% CI [+0.0005, +0.0048], $p=0.01586$) aur summer (mean change +0.00711, $n=169$, $p<0.000001$; DiD +0.00592, 95% CI [+0.0025, +0.0093], $p=0.00059$). Dono windows sign aur roughly size mein sehat hain, aur ye result ek from-scratch manual OLS-replication aur statsmodels-based primary-model ke beech kayi significant figures tak reproduce hota hai (Development Log Entry 26). Ye is study ke

pehle do reversals (Sections 4.2, 4.6) jaisa extraction-completeness ya control-contamination artifact nahi hai — Dynamic World ke image-count columns poori extraction ke dauraan fully populated rahe, SAR wale bug ke ulta. Ye absolute terms mein ek bahut chhota effect bhi hai — 'built'-probability ka teen-se-saat dus-hazaari-hissa — jo statistical significance sirf sample-size aur har village mein direction ki consistency ki wajah se clear karta hai. Iska size iska bada point nahi hai.

Do follow-up checks (Development Log Entry 27) is disagreement ko bina explain kiye examine karte hain. Pehla: ye koi outlier ya baseline-imbalance-artifact nahi hai. Top-5 aur bottom-5 change wale treated villages hataane se mean mushkil se hilita hai (full-year +0.00673 se +0.00621; summer +0.00711 se +0.00655), aur treated-vs-control baseline 'built'-probability dono windows mein statistically balanced hai (Mann-Whitney $p=0.996$ full-year, $p=0.155$ summer). Doosra, aur ye zyada dhyan-khichne wala hai: ye increase un treated villages mein concentrated hai jinka baseline 'built'-probability pehle se zyada tha (Spearman(built_before, built_change) = +0.272, $p=0.00001$, full-year). Control villages full-year mein aisa koi relationship nahi dikhate ($\rho=+0.020$, $p=0.585$), aur summer mein ulta, zyada ordinary pattern dikhate hain ($\rho=-0.229$, $p<0.00001$), jahan zyada-built-up control-villages kam hote hain — ek ordinary ceiling-effect jo ek bounded 0-se-1 scale par expected hai. Treated villages, ajeeb tarah se, wo ceiling-effect bilkul nahi dikhate.

Ek teesra check, is study ki apni history dekhte hue zaroori hai: sabhi 31 Sikkim treated villages Dynamic World ke summer window mein null wapas aate hain (0 mein se 31 valid), jabki full-year window mein sabhi 31 ek value dete hain, chahe teeno states mein sabse kamzor (+0.00098, Arunachal Pradesh ke +0.0075 aur Uttarakhand ke +0.0078 ke against). Yahi shape ka state-wide summer-dropout Development Log Entries 21-22 ne NDBI ke liye khoja aur fix kiya tha — is baar Dynamic World ke Sentinel-2 L1C pull mein, SR-Harmonized NDBI pipeline mein nahi, aur abhi tak check nahi hua ki ye ek monsoon-cloud-blackout hai ya ek aur archive-backfill-snapshot artifact, waisa hi jaisa Entry 22 ne resolve kiya tha. Isi wajah se summer Dynamic World DiD ($p=0.00133$ purane 732-list par) ek aise treated-sample par estimate hota hai jisme wo ek state missing hai jo full-year window mein sabse kamzor signal deta hai — jisse full-year result, jisme Sikkim shaamil hai aur phir bhi significant rehta hai, dono windows mein zyada bharosemand ban jaata hai.

Ek village-level check (Development Log Entry 32) is disagreement ko bina resolve kiye aur bhi saaf karta hai: individual-village level par, Dynamic World ka 'built'-change na to SAR se na NDBI se meaningfully correlate karta hai, kisi bhi window mein. Full-year: Dynamic World vs. SAR VV ($\rho=+0.063$, $p=0.319$, sirf 41% villages mein same sign), vs. SAR VH ($\rho=+0.077$, $p=0.224$, 37% same sign), vs. NDBI ($\rho=+0.102$, $p=0.109$, 37% same sign). In mein se kayi combinations mein, do proxies same disha sirf aadhe se kam villages mein point karte hain — ek coin-flip se bhi kam. Dynamic

World ka chhota, significant aggregate-DiD un hi villages se drive nahi ho raha jinhe SAR ya NDBI change-hua flag karte. Ye ek population-level statistical pattern hai, na ki koi village-by-village confirmation.

Ek aakhri check (Development Log Entry 42) — Dynamic World ka apna 2019-to-2021 pre-treatment placebo test — dono windows clean wapas aate hain (full-year $p=0.93406$, summer $p=0.17869$). NDBI/lights ke apne placebo-test ke ulta, jisne ek borderline exception flag ki thi (summer, $p=0.07421$, Section 7.5), Dynamic World kahin bhi koi pre-existing divergence nahi dikhata. Ye SAR/NDBI ke saath disagreement resolve nahi karta, lekin ek specific explanation ko rule-out zaroor karta hai: ye VVP-I se pehle wale kisi trend ka continuation nahi hai. Jo bhi isse paida hua, agar kuch paida hua, wo asli 2021-2025 treatment-window ke andar hi shuru hua, uske pehle nahi.

Section 3.12 ki primary/secondary hierarchy ke against padho: NDBI is study ka primary estimand hai. Dynamic World ka 'built' band kabhi nahi tha. Dynamic World ka role NDBI ko replace karna kabhi nahi tha, sirf SAR ke saath ek aur cross-check ki tarah baithna tha. In dono mein se, sirf SAR Section 7.1 ki asli maang ke tarah sensor-independent hai. Dynamic World NDBI ka sensor share karta hai, aur uske apne extraction-script ki documentation ke mutabik, Sentinel-2 ke L1C imagery-tier se aata hai, jabki NDBI Surface-Reflectance tier use karta hai. In do product-tiers ke beech atmospheric-correction ya cloud-handling ka farak ek plausible wajah hai is disagreement ki, saath hi ek simpler possibility ke saath ki ek deep-learning classifier ek fixed two-band index se zyada sensitive hai ek chhote change ke liye jo NDBI ya SAR ke coarser effective-resolution ko 500m-buffer scale par pakadne ke liye bahut subtle hai. Dono explanations abhi speculative hain. Inhe alag karna future-work ke liye chhoda gaya hai.

Section 5 — Discussion: Ye Study Khud Apne Baare Mein Kya Kehti Hai

Yahan ka main finding zyada tar physical-development ke upar-neeche jaane ke baare mein nahi hai. Ye hai ki is study ne do baar ek 'significant' result diya jo baad mein ek data-artifact nikla, koi solid effect nahi — aur dono baar jawab wahi raha: artifact ko fix karo aur jo bacha use report karo, result ko khade rehne mat do. Pehla tha summer-window built-up-area-increase, jo ek incomplete-extraction se aaya tha jisme Sikkim ke saare villages missing the. Ek baar wo extraction complete hone ke baad, aur treated/control data usi din pull hone ke baad, result har check ke under gayab ho gaya jo is study ne uspar phenka — control-group ke against, leave-one-district-out ke under, randomization-inference ke under, aur teeno buffer-radii ke across (Sections 4.2, 4.6, 4.8; Development Log Entries 21 se 22).

Doosra tha full-year night-lights control-group gap, jo un hi checks se ek poore development-cycle ke liye bacha raha (Entry 22 se aage), aur is study ka ek-akela-bacha-hua significant result report hua — jab tak Development Log Entry 23 ne nahi paaya ki jis 'non-VVP' control-group ke against ye compute hua tha wo clean nahi thi. 20 control-villages physically wahi settlements the jo treated villages, alag naam-spelling ya script mein, aur 3 aur officially listed VVP-I villages the unke listed, ungeocoded naamon se. Wo contamination properly hatne ke baad (control-list regenerate karke aur poori pipeline uske against re-run karke, Entry 25), result har specification par null ho gaya jo pehle significant paati thi: primary estimate, HC3 comparison, leave-one-district-out, randomization-inference, aur log1p-transformed check.

Isi complete summer-data mein ek naya result bhi dikhta hai jisne pehla reversal diya: built-up-change ab border/LAC se distance ke saath correlate karta hai (H3, Section 4.5), securitization theory ki prediction se ulti disha mein, aur ye paper-wide Holm-Bonferroni correction ko full-year lights-proximity result ke saath survive karta hai (Section 4.9). Ye ab usi leave-one-district-out aur randomization-inference checks se guzar chuka hai jinke under (ab-null) summer NDBI-change aur full-year lights-DiD dono fail hue the, aur dono mein zinda bachta hai, saath mein ek spatial-autocorrelation correction aur ek multi-year linearity/stability check jinse un pehle wale, retracted findings kabhi guzri hi nahi thi (Section 7.6, Section 6.11, Development Log Entries 27 aur 32). Ye ek dataset mein dikhna jisne ab do baar aisa result diya hai jo baad mein artifact nikla, khud ek yaad-dilaane wali baat hai: har surviving ya newly-significant result ko usi tarah check karo jaise is paper ne aakhir mein apne findings check kiye. Ek fresh finding ko bas isliye zyada believable maan lena galat hoga ki wo naya hai — isne apni jagah is discussion ke conclusions mein wahi scrutiny survive karke kamaayi, usse exempt hoke nahi.

History Woven In — 2026-09-18 Ka Ek Aur Zaroori Correction

Ye poora paragraph ek pehle version mein kehta tha ki ye result 'un checks se guzra hi nahi' aur 'ek open finding ki tarah report kiya gaya hai future robustness-pass ke liye' — seedha Section 7.6 ke bilkul ulta, jo is paper ke development mein baad mein add hui aur exact ulta kehti hai: ki teeno checks chal chuke hain aur result unhe survive karta hai. Ye paragraph kabhi update nahi hua jab Section 7.6 likhi gayi. Ek independent review ne is contradiction ko pakda; Section 7.6 hi sahi, current status thi, aur ye paragraph ab uske saath line mein laaya gaya hai. Ye is poore project ki wahi seekh hai jo hum baar-baar dekh chuke hain: ek lamba, kayi mahino tak badalta document apni khud ki purani baaton se aage nikal sakta hai, bina kisi ko batae — isliye external review, aur 'fresh-read' ka wahi rule jo ye teaching-PDF-series khud follow karti hai, sirf ek formality nahi, ek genuine zaroorat hai.

Isme se kuch bhi is claim ko support nahi karta ki VVP-I ke investment ne uske budget ke hisaab se ya uski scale ke hisaab se development produce kiya, ya ki uska rollout border-proximity ko asli zaroorat se zyada priority deta tha, ya, ab, complete, decontaminated data par, ki usne is specific 2021-2025 window mein koi bhi satellite-detectable change produce kiya — do compositing-windows se check kiya gaya, do proxies ke across (NDBI, night-lights), ek control-group ke against, aur teen buffer-radii par. Ye is paper ke dono pehle versions se ek zyada narrow conclusion hai, aur ek zyada defensible bhi: ek uniform null, kayi alag tareekon se check kiya gaya, do baar confirm kiya gaya un do data-defects ko theek se hataakar jinhone har baar ek 'surviving' result diya tha thodi der ke liye. Programme ka khud kabhi koi official impact-assessment nahi hua, uska parliamentary record khud isi gap ko accountability mein admit karta hai, aur kayi tareekon se kiya gaya ek satellite-audit jo koi detectable ground-level signal nahi paata, us gap ka ek solid jawab hai, ek khaali jagah se zyada. Ye khud bhi, apne aap mein, ek caution hai ki aisi study kaise ki jaati hai: wahi script-bugs, archive-timing-sensitivity, aur control-group-contamination jinhone is paper ke original, ab-retracted headline findings diye, wo utni hi aasani se kisi bhi comparable study mein ek false-positive de sakte hain jo checking band kar de, ya jo apni checking se mile problems ko fix karna chhod de, kaam poora hone se pehle.

Section 6 — Limitations: Jo Kamzoriyan Khud Disclose Ki Gayi Hain

6.1 — Village Coverage Gaps

Ladakh ke 35 sanctioned villages पूरी तरह village-level analysis से बाहर हैं — कौन से publicly-indexed source उनके नाम report नहीं करता, और इस gap को band करने के लिए एक RTI-request चाहिए होती जो इस study ने नहीं की। Himachal Pradesh सिर्फ एक छोटे illustrative-cases के set की तरह दिखता है, सिर्फ 7 में से 51 inhabited priority villages, और main statistical sample में नहीं बैठता। Uttarakhand के Pithoragarh block में 19 villages के confirmed नाम हैं लेकिन कौन से block assign नहीं है।

6.2 — Border Geometry Ek Cartographic Approximation Hai

Is study का हर distance-to-border figure Natural Earth की istemaal की हुई cartographic boundary-line पर तिका है — एक simplified version उस Line of Actual Control का जिसका alignment internationally agreed ही नहीं है। ये authoritative numbers के stand-in हैं। ये खुद authoritative numbers नहीं हैं। इनसे सिर्फ एक comparison की तरह पढ़ो, और कुछ नहीं।

Ek related, zyada specific point: 'distance to border/LAC' ka matlab hai Natural Earth ke data mein sabse *nazdeeki* India-related boundary-segment ki distance, specifically China-boundary nahi. Core sample ke against seedha check kiya: 251 mein se 220 villages (88%) sachmuch China-boundary ke sabse nazdeek hain, lekin 19 (zyaadatar Tawang aur West Kameng, Arunachal Pradesh) Bhutan ke sabse nazdeek hain, 8 (North Sikkim aur Pithoragarh, Uttarakhand) Nepal ke, aur 4 (Anjaw, Arunachal Pradesh) Myanmar ke. Us 12% sample ke liye, H3 ek alag desh ki frontier ki proximity measure kar raha hai, specifically LAC nahi. Ye phir bhi ek general border-securitization-hypothesis ke liye defensible hai — ye border-villages hain us wider sense mein jo VVP-I khud use karta hai — lekin ye un villages ke liye literally ek LAC-proximity-test nahi hai. H3 ke pehle se hi modest, zyaadatar null results (Section 4.5) ke saath, aur ek disputed line ke liye ek cartographic proxy par iski reliance (Section 6.2) ke saath, ye ek aur wajah hai H3 ko kisi bhi disha mein ek clean confirmatory result ki tarah na lene ki.

6.3 — Budget Correlation Ka Scope Aur Ecological Inference (RQ2)

EXAMPLE — Ecological Inference Kya Hota Hai

Socho tum ek state ka total electricity-bill dekh kar ye guess karna chahte ho ki ek particular mohalle mein kitni bijli use hoti hai — lekin tumhare paas sirf poore state ka total number hai, us mohalle ka alag se nahi. Agar tum seedha state ke total ko us mohalle ke against compare karo, to ye ek galat comparison hai, kyunki state ka total us mohalle ke alawa bhi bahut kuch cover karta hai. Yahi is study ke saath ho raha hai: budget figure poore state ke VVP-I portfolio ko cover karta hai, sirf un villages ko nahi jo is study ne geocode kiye. Ek 'group-level' number ko ek 'sample-level' number ke against seedha compare karna — isi ko 'ecological inference' ki galti kaha jaata hai.

RQ2: teeno core states ke paas ab valid summer-window data hai (Section 4.4, pehle sirf do the, Sikkim ka data wapaa aane se pehle), lekin teen states abhi bhi ek confirmatory test chalane ke liye kaafi kam hain, yahi wajah hai ki results hamesha description ki tarah dikhaye jaate hain, confirmation ki tarah kabhi nahi. Isse like-for-like test mat maano. Isse suggestive maano, aur note karo ki iski disha dono compositing windows ke beech palat jaati hai (Section 4.4) — ek aur wajah ispar zyada bharosa na karne ki.

6.4 — Compositing Window Sensitivity: Ek Resolved, Aur Ek Naya Khula Sawaal

Full-year aur summer-matched built-up-area-change ab seemat hain (dono null, Section 4.2), isliye ye jodi ab window-sensitive nahi hai jaisa is paper ke pehle versions kehte the. Ek khula sawaal pehle yahan flag hua tha: kya summer window ka 'Sikkim ka zero valid data' result ek permanent fact tha, ya bas Sentinel-2 archive ka us

din ka ek snapshot? Ab resolve ho gaya hai. Wo doosra tha (Development Log Entries 21-22). Ek naya khula sawaal uski jagah leta hai, usi complete re-extraction se aata hua: built-up-change ab border/LAC se distance ke saath summer window mein correlate karta hai (H3, Section 4.5), ek test jo is refresh se pehle bhi aur full-year window mein (jise ye refresh chhu nahi paaya) abhi bhi ek stable null hai. Kya ye Sentinel-2 ke usi archive-timing-effect ke against sensitive hai jisne pehle wale finding ko drive kiya, ye abhi tak check nahi hua. Lekin ye 2021/2023/2025 ke teen-point extraction ke against cross-check ho chuka hai (Section 7.6): iski linearity theek tarah tikti hai, iski temporal-stability nahi. Ye har specific test jo ab tak chalaya gaya use survive karta hai. Lekin 'jo test hua use survive karna' ek narrower claim hai 'VVP-I villages ke baare mein ek stable fact' se.

6.5 — Geocoding Coverage Aur Selection Bias

Sirf 559 mein se 258 villages (46%) geocode hue, aur Arunachal Pradesh ne sabse bada hit liya — uske 455 mein se 269 villages bahar gir gaye (Section 4.1). Ek village jise OSM/Bhuvan geocode nahi kar paaya, reasonably chhota, zyada remote, ya administrative lists mein kam achhi tarah record ho sakta hai — jo use kisi mushkil-se-pahunchne-wali jagah par hone ki zyada sambhavna dega. Ye ek fair guess hai, lekin isko bas assume nahi karna chahiye.

Village-size par ek Mann-Whitney U test Arunachal Pradesh par chalaya gaya, jahan population aur household counts har village ke liye maujood the chahe geocoding kaam kare ya na kare, aur koi significant farak nahi mila geocoded aur non-geocoded villages ke beech, round population mein (matched mean 135.0 vs. unmatched 145.7, $p = 0.310$) ya round households mein (matched mean 25.8 vs. unmatched 28.8, $p = 0.540$). Village-size se judi ek selection-bias yahan poori tarah rule-out nahi ho sakti, lekin kam-se-kam ye is relationship mein saaf-saaf nahi dikhti.

Selection-bias jahan dikhta hai wo village-size nahi hai. District hai. Wahi geocoded-vs-not comparison ko ek chi-square test ki tarah district-composition par chalao, size par Mann-Whitney ke bajaye, aur teeno core states mein ek bilkul alag kahani milti hai: Arunachal Pradesh (chi-square = 139.16, $p = 6.25 \times 10^{-25}$, 11 districts ke across jinke paas ek official village-list hai), Sikkim (chi-square = 6.01, $p = 0.0143$, 2 districts), aur Uttarakhand (chi-square = 6.24, $p = 0.0442$, 3 districts). Geocoding-success ek state ke districts mein barabar nahi phaila hua — ye kuch mein concentrated hai aur kuch mein lagbhag gayab hai.

District	State	Geocoding Rate
Kurung Kumey	Arunachal Pradesh	83.3% (70/84)
Dibang Valley	Arunachal Pradesh	72.0% (18/25)

West Kameng	Arunachal Pradesh	18.2%
Tawang	Arunachal Pradesh	16.8% (26/155)
Shiyomi	Arunachal Pradesh	0% (0/30)
Uttarkashi	Uttarakhand	100% (10/10)
Pithoragarh	Uttarakhand	59.3% (16/27)

Ye ek district-level pattern hai jise is study ka apna district-fixed-effects DiD design (Section 3.7) partially by-construction absorb karta hai, kyunki ye within-district compare karta hai, poore official village-universe ke across nahi — lekin iska matlab hai ki 251-village core sample bahut sambhavna hai ki un districts ko over-represent kare jinhe OSM/Bhuvan ki geocoding-pipeline achhi tarah cover karti hai, aur Tawang, Shiyomi, aur West Kameng ko under-represent kare specifically — Arunachal Pradesh ke sabse remote, sabse border-adjacent districts mein se teen. Agar geocoding-failure exactly usi tarah ki remoteness se correlate karta hai jise is study ka apna border-proximity hypothesis (H3) parwaah karta hai, to core sample har district ke andar zyada accessible, zyada representative nahi, villages ki taraf skewed ho sakta hai. Ye retroactively fix nahi ho sakta bina 301 officially-listed, currently-ungeocoded villages par ek fresh geocoding-pass ke, isliye ise ek open limitation ki tarah yahan disclose kiya gaya hai, kuch aisa nahi jise ye analysis resolve kar sake.

6.6 — Ground Truth Validation, Ab Ho Chuki Hai, Aur Ye Null Ko Reframe Karti Hai

Is section ka ek pehla version kehta tha ki koi ground-truth positive-control check karne ke liye nahi hai — koi independently confirmed, dated, named, completed VVP-I project is study ke 258-village geocoded sample ke andar nahi girta tha. Wo gap ab band ho gayi hai (Section 7.3, Development Log Entry 28). Teen aise villages mile: Walong, Kaho, aur Musai, sab Anjaw district, Arunachal Pradesh mein. Har proxy ko in teeno ke against ek-ek karke check karna, is study ka sabse important interpretive-finding deta hai, headline null ke alawa. VIIRS night-lights confirmed nayi infrastructure ko saaf pakadti hai — Walong ka radiance lagbhag double hua (0.89 se 1.80, full-year) aur Kaho ka 82% badha (0.51 se 0.93). NDBI, wo proxy jo primary estimand ke peeche hai, teeno villages mein *ulti disha* mein move hua (Kaho -0.057, Musai -0.064, Walong -0.038, full-year), chahe har ek mein confirmed nayi construction hui thi isi exact period mein. Ye Section 4.2 ke pooled null ko overturn nahi karta — teen villages ek case-study hain, ek sample nahi — lekin ye is study ka pehla solid saboot hai ki NDBI VVP-I ke funded development ko systematically kyun miss kar sakta hai. Ye ek asli saboot hai, sirf ek plausible-lagne-wala guess nahi: chhote, point-scale additions jaise ek solar streetlight, ek hostel building, ek basketball court, ek 500m buffer area-average mein wash-out ho sakte hain, jabki ek radiance-sensor jo

point light-sources pakadne ke liye bana hai, unhe seedha pakad leta hai.

6.7 — Control-Group Comparison Mein Baseline Imbalance

Is paper ke pehle versions treated aur control villages ko 2021 baseline par teen mein se chaar outcome/window combinations mein significantly alag report karte the, phir, Development Log Entry 22 se aage, sirf ek mein: full-year window ka night-lights baseline. Wo char mein se sabse bada imbalance tha, aur us par hi is study ka ek-bacha-hua control-group DiD result tika hua tha. Ek baar control-group-contamination Entry 23 ne pakda aur live data ke against theek se fix ho gaya (Entry 25) — 20 control-villages hataakar jo treated villages ke exact coordinate-duplicates the aur 3 aur jo ek official-lekin-ungeocoded priority-village se naam-match karte the — full-year night-lights baseline significantly-imbalanced rehna band ho gaya: $p = 0.08976$. Chaaron baseline-balance combinations, thodi der ke liye, statistically indistinguishable the treated aur control villages ke beech 2021 mein.

Ye tika nahi. Ek aur, unrelated correction — is section ke saathi-table (Section 6.10) ne jo district-verification-gap flag ki thi use band karna, Development Log Entry 37 — ne control-list ko regenerate kiya (732 se 721 villages, Tawang aur Pithoragarh ab boundary-verified, bbox-matched nahi) aur exactly yahi imbalance dobara khol diya: treated mean 0.42821 vs. control 0.43493, $n = 251/721$, $p = 0.03558$. Baaki teen combinations corrected list par balanced rehti hain. Ye seedha disclose kiya gaya hai, chhupaya nahi: ek asli, independently-verified data-quality-problem (unverified district-labels) ko fix karne se ek baseline-imbalance khula jo ek pehli, kam-sahi control-list ke saath nahi dikh raha tha. Ye apne aap se prove nahi karta ki treated aur control villages bina treatment ke identically move karte (Section 7.5 dekhati hai ki ek asli pre-treatment-panel kya aur add karti), aur iska matlab hai ki full-year night-lights DiD result — jo corrected list par null rehta hai ($p = 0.45334$, Section 4.6) — ab is ek outcome/window combination ke liye ek well-matched baseline par khada nahi maana ja sakta. Null khud reverse nahi hua — ek imbalanced baseline par baithe null ka matlab kya hai, ye ek genuinely alag, kamzor claim hai us same null se jo ek balanced baseline par baitha ho, aur ye paper us farak ko report karta hai, use chhupata nahi.

6.8 — Non-Monotonic Multi-Year Pattern

Pooled 2021-vs-2025 summer NDBI comparison apne aap mein ek null hai ab (Section 4.2), isliye teen-point trend ke liye explain-away karne layak koi significant headline-expansion bacha nahi hai. Lekin uske andar ka sub-period pattern abhi bhi tikta hai, aur is paper ke data-refresh se untouched hai: ek decline 2021 se 2023 tak (mean change -0.0231, akele significant nahi), phir ek significant recovery 2023 se 2025 tak (mean change +0.0222, $p < 0.000001$), jo net-out hoke ek non-significant teen-saal-ka-trend banata hai (Wilcoxon $p = 0.442$, panel regression $p = 0.728$). Ye Section 4.2 ke ab-null do-point-result ko match karta hai, contradict nahi karta.

6.9 — Buffer Radius Comparison Mein Archive-Timing Confound, Resolved

Is section ka ek pehla version ek khula gap flag karta tha: Section 4.8 ke peeche ki 250m aur 1km buffer-extractions 500m primary-run se ek alag din pull hui thi, isliye teeno principle mein scene-counts ya composite-values par disagree kar sakte the, sirf isliye kyunki Sentinel-2 ka archive purani dates ke liye scenes backfill karta rehta hai, buffer-radius se bilkul alag. Wo gap ab buffer-sweep ke liye bhi band hai: teeno radii (250m, 500m, 1km) ek hi din back-to-back re-extract hue, poore 251-village core sample ke liye (Development Log Entry 31). Result teeno par ek saaf null rehta hai, aur timing-gap band karne se numbers mushkil se hile. Ye is study ke archive-timing-issues wale poore family ka aakhri khula item band karta hai, Entries 21-22 ke primary comparison ke same-day-fix ke saath.

6.10 — Summary: Threats To Validity

Sections 3, 4, aur 6 in sabko ek-ek karke pehle hi discuss kar chuke hain, wahin jahan har ek sabse relevant hai. Ye table bas unhe ek saath rakh deti hai, taaki ek reader poori tasveer dekh sake bina dus alag paragraphs se piece-together kiye. Neeche ek condensed version hai — poori detail upar diye gaye sections mein hai.

Threat	Status
Cloud contamination (Sentinel-2)	Tested — bug mila (SCL snow-bug), fix hua, ab QA60/SCL sehamat, dono null (Section 6.12)
Snow contamination (high-altitude)	Mitigated by design — season-matched window snow ko avoid karti hai (Section 3.5)
Archive timing inconsistency	Resolved — primary comparison aur buffer-sweep, dono same-day re-extracted (Section 6.9)
Missing villages / geocoding (46%)	Partially tested — size se no-bias (AP), district se strong bias (Section 6.5)
Control baseline imbalance	Mixed — 3/4 balanced, full-year lights resolved phir dobara imbalanced (Section 6.7)
True parallel pre-trends	Resolved — 3/4 clean placebo, summer NDBI borderline (Section 7.5)
Few (14) district clusters	Tested — HC3, leave-one-out, randomization, wild-cluster-bootstrap, sab null (Section 4.6)

Spatial autocorrelation	Resolved — confirmed present, spatial-correction applied (Section 6.11)
Border/LAC geometry as legal claim	Addressed by framing — cartographic/comparative only (Section 6.2)
Control selection distance metric	Resolved — geodesic method, fixed in code (Development Log Entry 24)
Control group contamination	Resolved — 0 coordinate-overlaps, 0 name-matches (Development Log Entry 37)
Control district-verification gap	Resolved for 2/3 districts — Sikkim North still bbox-only (Development Log Entry 37)
Differential imaging density	Tested and cleared ($\rho=0.038$, $p=0.551$)
Treatment not randomly assigned	Addressed by framing — inherent limitation of observational design
NDBI as development proxy	Tested — correlates with building-counts ($\rho=0.399$), weakest of three proxies (Section 7.2)
No ground truth validation	Resolved — Walong/Kaho/Musai case study, NDBI wrong-direction all three (Section 6.6)
Multiple testing (8 p-values)	Applied — Holm-Bonferroni, scoped to H1/H3 family only (Section 4.9)
Single sensor/algorithm dependency	Partially resolved — SAR confirms null, Dynamic World disagrees (Section 4.10)
VIIRS resolution vs. buffer	Disclosed as sensor-resolution limit, not fixable by more data

6.11 — Spatial Autocorrelation: Confirmed Present Aur Significant, Ab Corrected

Section 6.10 ki table pehle spatial-autocorrelation ko 'identified but not tested' ki tarah list karti thi. Ab test ho chuka hai (Development Log Entry 27), ek saaf jawab ke saath: ye maujood hai. Moran's I seedha compute hua ($k=8$ nearest-neighbor weights, haversine distance se, permutation-based significance 999 label-shuffles se). Summer NDBI change strong positive clustering dikhata hai (Moran's $I = 0.346$, permutation-null mean -0.004 , SD 0.028 ; $p = 0.001$) — paas-paas ke villages ke changes ek-dosre se kaafi zyada milte-julte dikhte hain, chance se zyada. Full-year lights change wahi pattern dikhata hai, kamzor ($I = 0.076$, $p = 0.003$). Ye sirf H3 ki distance-relationship dobara nahi bata raha — distance-se-uske-linear-trend hataane

ke baad bhi, Moran's I = 0.272, p = 0.001 rehta hai.

Iska practical matlab: is study ke 251 core-sample villages utne statistically-independent observations nahi hain jitna har test ka stated significance-level chupke se maan leta hai. Ye correction ab nikaali ja chuki hai (Development Log Entry 32): ek simultaneous-autoregressive parameter usi k=8 weight-matrix par calibrate hua taaki synthetic fields observed Moran's I reproduce karein, phir 2,000 aise spatially-autocorrelated-lekin-otherwise-random synthetic fields ko distance-to-border values ke against correlate kiya gaya, ye dekhne ke liye ki sirf spatial-structure wala ek field, border-distance se koi relationship na rakhte hue, kitni baar ek observed-jaisi extreme correlation deta hai.

H3 Finding	Naive p	Spatially-Corrected p
Summer NDBI proximity (rho=0.291)	0.0000026	0.02949 — bilkul kinare par
Full-year lights proximity (rho=-0.252)	0.0000548	0.00350 — comfortably significant

Dono surviving H3 correlations correction ke baad bhi significant rehte hain, lekin ek ke liye margin doosre se kaafi chhota hai. Section 7.6 ki linearity aur multi-year findings ke saath combine karke, summer NDBI proximity result is study ke do Holm-significant H3 findings mein se zyada fragile hai, har axis par jo ab tak check hua hai.

6.12 — Cloud Masking Sensitivity: SCL Vs. QA60, Chalaya, Ek Bug Mila, Fix Hua, Ab Sehmat

Section 6.10 ki table pehle SCL-band cross-check ko 'prepared but not executed' ki tarah list karti thi. Ye chal chuki hai (Development Log Entry 33), SCL classes 2/4/5/6/11 (dark-area, vegetation, bare-soils, water, snow) ko 'clear' maankar — aur ye pehli baar ek unresolved-instability jaisi dikhi: SCL mask sirf 200 mein se 258 villages ke liye valid data deti thi, aur un 200 par ek significant positive summer NDBI change (mean +0.01169, p = 0.0000045) deti thi, ek QA60-null (p = 0.449) ke against, sirf moderate village-level agreement ke saath dono masks ke beech (rho = 0.544).

Us check mein ek bug tha. Is repository ke ek external review ne ise pakda, is project ke apne kisi bhi test ne nahi: Sentinel-2 Scene Classification Layer mein class 11 snow/ice hai, clear ground nahi, aur original class-list ise 'clear' ki tarah guzar rahi thi. June-September composites ke liye Himalayan border-villages par, ek 500m buffer ke andar snow-cover ek possibility hai jise seriously lena zaroori hai, aur snow-contaminated pixels par compute hui NDBI built-up-ness bilkul measure hi nahi karti. `extract_scl_cloud_mask_ndbi.py` fix hua `SCL_CLEAR_CLASSES = [2, 4, 5, 6]` tak

(snow hataakar), ek stricter [4, 5, 6] variant bhi ek doosre cross-check ki tarah extract hui, aur dono live Earth Engine ke against rerun hui (Development Log Entry 34).

Result badalta hai. Snow ko sahi tarah exclude karne ke baad, SCL-masked coverage 251 mein se 258 villages tak badhti hai, ab QA60 ki coverage se lagbhag exactly match karti hai aur ab Arunachal Pradesh ki taraf skewed nahi hai, aur un 251 villages par summer NDBI change ab significant nahi hai (mean +0.00112, Wilcoxon $p = 0.28797$). QA60 identical 251 villages par null hi rehta hai (mean -0.00090, $p = 0.43583$). Dono masks ab sehmat hain, dono null. Unke beech village-level agreement bhi kaafi zyada hai pehle wale se (Spearman $\rho = 0.734$, pre-fix $\rho = 0.544$ ke against) — jo fit baithta hai is baat se ki wo pehla 'moderate agreement' figure khud ek snow-affected, non-representative subset ke against masks compare karne ka artifact tha.

Ek stricter SCL variant (classes [4, 5, 6], dark-area pixels bhi hataakar) bhi extract hui, ek aur cross-check ki tarah, sirf pehle attempt ka bug-fix nahi. Ye baaki dono se sehmat hai: $n = 251$, mean change +0.00187, Wilcoxon $p = 0.17501$, QA60 ke saath village-level agreement $\rho = 0.736$. Teen masks, teeno sehmat nulls — isliye conclusion is baat par nahi tikti ki clear/not-clear ki line exactly kahan khinchi jaaye, ek defensible range ke andar; sirf snow-exclusion par tikti hai.

Ek loose-end is rerun se flag karna zaroori hai, explain-away nahi karna: SCL ki valid coverage 200/258 se 251/258 tak badhi class-fix ke baad — jo ulta hai us baat se jo ek 'clear' class hataane se mechanically hona chahiye, kam clear-classes ka matlab hona chahiye barabar ya kam coverage, zyada nahi. Sabse plausible explanation, is project ki archive-backfill history dekhte hue (Development Log Entries 21-22, jahan ek poore state ki summer-imagery shuru mein missing thi aur baad mein available ho gayi), ye hai ki original SCL-extraction aur is rerun ke beech collection mein zyada scenes aayin, class-list-change se alag. Ye seedha confirm nahi hua, isliye ise ek open question ki tarah record kiya gaya hai, sach maan kar nahi.

Is section ka asli matlab ye nahi hai ki 'null ab doubly confirmed hai aur kehne ko kuch nahi bacha.' Ye hai: is study ke ek external review ne ek coding-error pakdi ek check mein jise study ne khud ek khuli sensitivity ki tarah flag kiya tha, aur fix ho gaya aur rerun hua. Koi bas uspar behas nahi karta raha. Pichhle version mein report hui specific-instability wahi bug nikli. Cloud-contamination measure karne ke tareeke ko lekar kabhi koi asli disagreement thi hi nahi. Primary NDBI null ab dono cloud-masking-conventions ke under tikta hai, ek baar mask sahi tarah define hone ke baad. Ye Dynamic World wali alag, abhi-bhi-khuli tension ko chhoota nahi (Section 4.10) — wo disagreement kabhi cloud-masking ke baare mein thi hi nahi, aur is fix se unaffected rehti hai.

Section 7 — Future Work: Jo Zyaadatar Ab 'Future' Nahi Raha

Neeche diye gaye items is study ke review-process se nikle specific extensions hain. Inhe Discussion aur Limitations se jaan-bujh kar alag rakha gaya hai, taaki jo ho chuka hai wo jo abhi sirf ek possibility hai, uske saath mix na ho. Ek dilchasp baat: in saato mein se paanch already ho chuke hain — is section ka naam 'Future Work' hai, lekin iska zyaadatar hissa ab past-tense mein likha ja sakta hai.

7.1 — SAR-Based Change Detection, Poori Ho Chuki (Section 4.10 Dekho)

Is paper ke pehle versions ise future-work ki tarah list karte the: Sentinel-1 SAR (VV/VH backscatter) ek fully sensor-independent cross-check ki tarah, NDBI-based primary-estimand par chalaya jaaye, cloud-cover aur optical-snow-contamination confound ko bypass karte hue jo is study ke compositing-window-check ki asli wajah thi. Ye apne aap 'dispute resolve' nahi karega full-year aur summer-matched optical windows ke beech. Jo ye deta hai uski jagah hai ek asli, independent measurement jise kisi bhi window ke known-artifacts chhoo nahi sakte. Ye ab chal chuka hai, treated aur control villages ke liye, dono compositing windows mein, aur poori tarah Section 4.10 mein likha hai: SAR dono bands aur dono windows mein null hai, is study ke primary conclusion ke liye sabse mazboot confirmation.

7.2 — Building Footprint Ya Sub-500m Structural Analysis, Poori Ho Chuki

500m NDBI buffer built-up surface, bare-rock, farmland, aur natural-vegetation ko ek area-averaged number mein daal deta hai — ek coarse tareeka ye pakadne ka ki VVP-I village-level par kya invest kar raha hai: individual roads, buildings, chhoti installations. Ye section originally is check ko ek independent building-footprint-dataset ke against proposal karta tha, aur wo check ab chal chuka hai (Development Log Entry 30). Google ka Open Buildings dataset, jo yahan use hua, ek single current-vintage snapshot hai, multi-temporal series nahi — isliye koi 2021-vs-2025 footprint-comparison nahi chal sakti, sirf ek cross-sectional 2025-era ('after') values ke against.

Proxy	Rho Vs. Building-Footprint Count
NDBI	0.399
Night-lights	0.478
Dynamic World 'built'	0.808

Building-footprint-count NDBI ke saath $\rho=0.399$ par correlate karta hai, night-lights ke saath $\rho=0.478$ par — dono moderate, kaafi hadd tak expected. Dynamic World ke 'built' probability ke saath ye $\rho=0.808$ par correlate karta hai, NDBI ke number se lagbhag double, aur teeno mein sabse mazboot. Ek bilkul independent, non-satellite ground-truth ke against, Dynamic World ki reading is ground-truth ko NDBI ki reading se kaafi zyada zyada closely track karti hai. Ye Section 4.10 ke Dynamic-World-vs-NDBI/SAR disagreement ko *change ki disha* par resolve nahi karta — ek strong cross-sectional level-correlation kuch nahi kehti is baare mein ki koi proxy *change* sahi tarah detect karta hai ya nahi — lekin ye ek wajah hai Dynamic World ki reading ko zyada seriously lene ki, us simple 'ye primary-estimand se disagree karta hai isliye galat hoga' framing se zyada.

Section 7.3 ke teeno case-study villages yahi cheez seedha, footprint-basis par dikhate hain. Walong ke paas 524 detected buildings hain (119,779 sq m total footprint area), ek substantial settlement, sample-wide median 34 ke against. Kaho ke paas 78 aur Musai ke paas 108 hain, dono us median se kaafi upar. Inme se koi bhi ek chhota hamlet nahi hai — jo khud informative hai: Walong ka confirmed naya terminal-building aur hostel, ek pehle-se-substantial built-environment mein additions hain, exactly wo kism ki addition jise ek 500m area-average sabse kam register karega ek badi percentage-change ki tarah.

7.3 — Ground Truth Positive-Control Validation, Poori Ho Chuki

Is section ka ek pehla version kehta tha ki 258-village geocoding-sample ke andar koi known, dated, independently-verified completed VVP-I initiative nahi mila. Wo search ab dobara hui, aur is baar kaam kiya (Development Log Entry 28). Anjaw district, Arunachal Pradesh ke ek official do-din ke tour, state ki Deputy Chief Minister dwara, ki news-coverage (Arunachal Times, arunachalobserver.org, 31 October se 1 November 2023) 15 specific projects document karti hai kayi Anjaw villages ke across, jin mein se teen is study ke 258-village sample mein hain: Walong (ek civil terminal-building Walong ke advanced-landing-ground par, aur ek 30-bed girls' hostel Walong Government Secondary School mein), Kaho (ek basketball court, solar street-lights, aur ek school library, explicitly 'under Vibrant Village Program' funded report hui), aur Musai (solar street-lights aur doosri 'model village' facilities, wo bhi explicitly VVP-attributed). Sabhi inaugurations is study ke 2021-se-2025 comparison-window ke andar aati hain, ek specific date ke saath (31 October se 1 November 2023) — exactly wo 'specific, dated, named' positive-control case jise is section ka pehla version dhoondh nahi paaya tha.

Proxy	Kaho	Musai	Walong
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Night-lights (full-year)	+0.420 (82% rise)	+0.011	+0.909 (~double)
NDBI (full-year)	-0.057 (ulta)	-0.064 (ulta)	-0.038 (ulta)
Dynamic World 'built'	+0.044	+0.006	+0.016
SAR VV/VH	Noise-level, koi signal nahi	—	—

Ye ek 3-village case-study hai, ek statistical test nahi, aur isse usse zyada nahi padhna chahiye. Ye establish nahi kar sakta ki VIIRS 'sahi' proxy hai, ya NDBI general mein 'galat' hai. Lekin in limits ke andar, ye is study ke null ke ek hisse ke peeche ek specific aur useful explanation ke liye solid saboot hai. Ye sirf ek guess nahi hai: confirmed projects yahan — solar streetlights, ek hostel, ek basketball court, ek terminal-building — sab chhote, point-scale additions hain, exactly wahi 'individual roads, buildings, chhoti installations' jinhe Section 7.2 pehle hi NDBI ke 500m area-average ke liye poorly-suited flag kar chuka hai.

7.4 — Himachal Pradesh Aur Ladakh Ke Liye RTI Follow-Through

Is study ne jaan-bujh kar in do states ke missing village-wise annexures ke liye RTI-requests file na karne ka faisla kiya, jo BO_Development_Log.md mein record hai. ANALYSIS_FREEZE.md ki scope-boundary-note ke mutabik (September 2026 mein decide hua), ye ek disclosed, permanent scope-boundary maana jaata hai. Ye koi khula task nahi hai jo uthaya jaaye — aur kuch badle bina revisit nahi hoga, is project ke scope ke against cost aur multi-month reply-timeline dekhte hue.

7.5 — Control-Group Ke Liye Ek Asli Pre-Treatment Panel, Poori Ho Chuki

Ek asli pre-treatment panel pehle exist nahi karta tha, kyunki is study ka control-group comparison (Sections 3.7, 4.6, 6.7) hamesha sirf ek pre-period — 2021 — ke saath kaam karta tha. Section 6.7 ka level-balance-check confirm kar sakta tha ki treated aur control villages 2021 mein similar levels par shuru hue, lekin ye nahi ki wo treatment se *pehle* parallel move kar rahe the — exactly wo jo DiD design ki identifying-assumption ko chahiye. Wo gap ab band hai (Development Log Entry 30): ek doosra, pehla pre-treatment saal (2019, dono windows, dono groups) extract hua, aur ek placebo-DiD chalaya 2019-se-2021 ke change ko compare karte hue treated aur control villages ke beech — ek stretch jiske dauraan VVP-I ka koi possible asar ho hi nahi sakta tha, kyunki programme February 2023 tak sanction hi nahi hua tha.

Outcome / Window	Placebo DiD (2019→2021)	p-value
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Full-Year NDBI	+0.00398	0.73497 — clean
Full-Year Lights	-0.00647	0.75898 — clean
Summer Lights	+0.00632	0.52021 — clean
Summer NDBI	+0.01173	0.07421 — borderline (raw Mann-Whitney $p=0.00003$)

Teen mein se teen checks clean wapas aate hain, is study ke DiD-design-assumption ko mazbooti se support karte hue. Chautha ek flagged exception hai: summer NDBI ka placebo-DiD-coefficient district-fixed-effects specification ke under conventional threshold par significant nahi hai, lekin close hai, aur unadjusted raw Mann-Whitney comparison sharply significant wapas aata hai ($p=0.00003$). Seedha kaha jaaye to: district-composition zyaadatar, lekin poori tarah nahi, ek pre-existing farak explain karta hai ki treated aur control villages ka summer NDBI 2019 se 2021 ke beech kaise move hua. Ye ek borderline result hai, ek clean pass nahi, aur waisi hi report ki gayi hai. Zaroori baat: ye borderline pre-period-effect (+0.01173, treated trending up) ab-null summer NDBI DiD (-0.00275, Section 4.6) se *ulti* disha mein chalta hai — ye koi pehle-se-maujood trend nahi hai jo main-result ko explain-away kar raha ho.

7.6 — Naye Summer H3 Border-Proximity Result Ko Stress-Test Karna, Teeno Checks Ab Poore

Section 4.5 ki summer-window finding, ki built-up-change border/LAC se distance ke saath correlate karta hai ($\rho = 0.291$, $p = 0.0000026$), sirf isi paper ki extraction complete hone ke baad hi dikhi (Development Log Entry 22). Ye ab un teeno checks se guzar chuki hai jinke around wo pehla, ab-retracted, summer NDBI-change-result bana tha — sirf do nahi, teeno: leave-one-district-out (14 mein se 14 district-dropped subsamples mein significant, wahi sign throughout, ρ range [+0.183, +0.337], koi ek district ise drive nahi kar raha), aur randomization-inference (2,000 permutations) observed ρ ko 2,000 shuffled values se aage rakhta hai ($p<0.001$).

Yahi script doosre Holm-significant H3 result par bhi chala, full-year night-lights proximity ($\rho = -0.252$). Paper mein kahin aur (Sections 4.5, 4.9), ye result 'robust' isliye describe hota hai kyunki ye kayi data-re-extractions ke across 'hamesha se' significant raha — jo ek alag, kamzor claim hai sachmuch leave-one-out ya randomization-inference-checked hone se. Wo check ab tak nahi hua tha. Ab ho chuka hai: ye bhi survive karta hai (14/14 leave-one-out, $p<0.001$ randomization-inference).

Teesra item — linearity poori distance-range mein, versus ek end par concentration, plus 2021/2023/2025 teen-point extraction ke against ek seedha comparison — ye bhi ab poora hai (Development Log Entry 27), aur yahan results upar wale leave-one-out/randomization pairing se zyada mixed hain. Linearity par: summer NDBI

proximity correlation lagbhag ek seedhi line hai paanchon distance-quintiles ke across (nearest se farthest: mean change -0.0182, -0.0074, +0.0009, +0.0071, +0.0135 — ek progressive gradient, quadratic-term significant nahi), ek clean result. Full-year lights proximity correlation aisi nahi hai: wahi paanch bins dete hain +0.0815, +0.1557, +0.0090, +0.0073, +0.1765 — badhna, do baar girna, phir farthest bin mein wapas upar jump karna, ek monotonic gradient nahi.

Multi-year cross-check par: dono correlations independent teen-point extraction ke against closely reproduce hoti hain (summer NDBI $\rho=+0.293$ do-point extraction ke +0.291 ke against; full-year lights $\rho=-0.252$ -0.252 ke against exactly). Do sub-periods mein split karo, lekin, aur koi bhi correlation waqt ke saath stable nahi rehta. Summer NDBI ka relationship poori tarah sign palat leta hai: distance vs. 2021-to-2023 change $\rho=+0.426$ hai (headline-result jaisi hi disha, aur bhi mazboot), lekin distance vs. 2023-to-2025 change $\rho=-0.222$ hai (ulta sign, wo bhi significant). Reported positive-correlation poore period par, asal mein do oppositely-signed halves ka ek net-result hai — ek steady gradient nahi jo poore period mein chalta ho. Full-year lights zyada consistent hai, phir bhi uneven: 2021-to-2023 $\rho=-0.239$ (wahi disha) kamzor hokar 2023-to-2025 $\rho=-0.078$ (wahi disha, ab significant nahi) ho jaata hai.

In do naye findings mein se koi bhi H3 result ko overturn nahi karta. Dono valid, Holm-significant correlations rehte hain, aur dono leave-one-out/randomization-inference pairing ko abhi bhi pass karte hain. Lekin 'stress-tested aur surviving' ko ab ek zyada precise description chahiye jo pehle chahiye nahi thi: ek district-dropout aur ek label-permutation-check survive karna ek stable, linear, time-consistent relationship hone jaisa nahi hai. Us finer reading par, sirf summer NDBI proximity result ki linearity cleanly tiki hai, uski temporal-stability nahi, aur full-year lights proximity result ki linearity bhi nahi tiki.

7.7 — Statistical Power: Ye Design Kya Detect Kar Sakta Tha

EXAMPLE — Cohen's D Aur MDE Kya Hote Hain

Socho tumhare paas ek metal-detector hai, aur tum use ek maidaan mein kisi chhupe hue sikke ko dhundhne ke liye use karte ho. Agar sikka nahi milta, to do possible wajah hain: ya to wahan koi sikka hai hi nahi, ya tumhara detector itna sensitive nahi tha ki use pakad sake. Cohen's d, aur us se nikalne wala 'Minimum Detectable Effect' (MDE), batate hain tumhara detector kitna sensitive tha — matlab, sabse chhota effect kitna hona chahiye tha taaki tumhara test use 80% confidence ke saath pakad paata. Agar ye number chhota hai (matlab detector bahut sensitive hai), to ek null-result ka matlab hai 'wahan sachmuch kuch nahi tha', 'detector kamzor tha' nahi.

power_analysis.py H1 (treated-only, paired) aur H4 (control-group DiD) tests ke liye ye Minimum Detectable Effect (MDE) nikalta hai, conventional significance ($\alpha=0.05$) aur power (80%) par, is study ke degrees-of-freedom par t-quantiles use karte hue

($n-1=250$ H1 ke liye; 14 districts – 1 = 13 H4 ke liye), ek large-sample z-approximation ke bajaye. Sidedness har test ke apne design ke mutabik hai: H1 ka MDE one-sided hai, Wilcoxon signed-rank test ke apne alternative='greater' specification se match karte hue; H4 ka MDE two-sided hai, cluster-robust DiD model ke apne two-sided p-values se match karte hue.

H1 ka MDE, Cohen's d ki tarah, 0.157 par aata hai har outcome/window combination mein — ek design-ki-apni property, sirf n aur df par depend karti hai, outcome ke scale par nahi — Cohen's usual 0.2 threshold ('small' effect ke liye) se comfortably neeche. Seedha kaha jaaye to: is study ka primary design ek chhote-se-moderate effect ko pakadne ke liye kaafi sensitive tha agar wo exist karta, isliye H1 ka null kuch maayne rakhta hai, sirf 'kaafi power nahi thi' ka case nahi hai. H4 ka cluster-based design, jaisa expected tha, kam sensitive hai (MDE 0.01918/0.13983 raw units full-year NDBI/lights ke liye, 0.01331/0.11792 summer ke liye), lekin har observed H4 coefficient us threshold ke andar hi baithta hai — 15 se 35% respective MDE ka. Ye ek near-miss nahi hai. Ye ek saaf gap hai jo observe hua uske aur design kya pakad sakta tha uske beech.

Section 8 — Conclusion

Ye study conclude nahi kar sakti ki jitni development ek is size ke budget se expect hogi — ya us scheme ki apni goal, steady annual progress, chhodo — India ke Vibrant Villages Programme ke priority villages mein, paanch border states aur union territories ke across, sachmuch achieve hui hai. Apne primary measure par — NDBI aur night-lights, QA60-masked, treated-only comparison ke against check kiya, control-group ke against, har buffer-radius par test kiya, is study ke chhote (14) cluster-count ko address karne wale ek exact wild-cluster-bootstrap ke under, aur ek 2019-se-2021 parallel-pre-trends-placebo-test ke under — ye study conclude nahi kar sakti ki koi satellite-detectable effect bilkul bhi exist karta hai. Ye ek conclusion ka settled version hai jo is study ne do baar pahunchaa, har baar ek problem dhoondh kar aur fix karke, kabhi bas jawab assume karke nahi.

Pehla: ek pehla version ek significant summer-window built-up-area-increase report karta tha jo har robustness-check survive karta tha jo uspar phenka gaya. Wo result ek incomplete-extraction par tika tha jisme Sikkim ke saare 31 villages missing the, aur ek baar wo extraction complete hone ke baad, ye ek saaf null tak palat gaya (Sections 4.2, 4.6, 4.8; Development Log Entries 21 se 22). Doosra: full-year night-lights control-group gap jisne us result ki jagah li, ek contaminated control-group par tika hua nikla, aur wo fix hone ke baad, wo bhi har specification par null ho gaya (Development Log Entries 23 se 25).

Do cheezein ab is null ke saath baithti hain, aur wo ek-dosre jaisi cheez nahi hain. Ek border-proximity-correlation jise ye paper kabhi naya aur un-checked flag karta tha, ab leave-one-district-out, randomization-inference, aur ek spatial-autocorrelation-corrected permutation-test se guzar chuki hai, aur teeno survive karti hai (Sections 6.11, 7.6) — ek seedha secondary finding, abhi bhi correlational, securitization-theory ki prediction se ulti disha mein chalti hui. Aur primary-measure ko do satellite-proxies ke against triangulate karna, null ko simply confirm nahi karta: Sentinel-1 SAR, jo sachmuch sensor-independent hai, isse sehmat hai, lekin Dynamic World ka built-up-classifier dono windows mein ek significant control-group-effect dikhata hai, building-footprint ground-truth ke against NDBI se zyada mazbooti se validate karta hai, aur is study ke teeno confirmed construction-case-studies ko seedha miss karta hai jinhe Dynamic World aur night-lights correctly detect karte hain. Ye ek asli, unresolved tension hai, koi bug nahi.

Ek teesra item is section ke ek pehle version mein ek dosri khuli tension jaisa dikhta tha — QA60-vs-SCL cloud-mask cross-check jo summer NDBI result ko null se significant mein palatta dikha. Ye external review par, ek coding-bug nikla khud SCL mask mein — snow pixels 'clear' ki tarah gine ja rahe the. Fix karke aur rerun karke, dono masks ab sehmat hain, dono null (Section 6.12, Development Log Entries 33 se 34). Ye teesra item yahan ek resolved-bug ki tarah report hota hai, teesri khuli finding ki tarah nahi, aur readers ko ise waise hi weigh karna chahiye: ye is project ke review-process ke ek error pakadne ke baare mein kuch kehta hai. Ye ye nahi kehta ki underlying science unsettled hai.

Isme se kuch bhi primary finding ko reverse nahi karta jaise wo do pehle artifacts kar chuke — Sikkim ki missing imagery, contaminated control-group. Wo dono ek data-problem nikle, aur fix hone ke baad, poori tarah gayab ho gaye, aur abhi-bataya-hua cloud-mask-bug isi pattern ka ek teesra case hai — ek coding-error jo ek finding jaisa dikha jab tak fix nahi hua. Triangulation jo dikhata hai, wo ek bug nahi hai: is study ka primary measurement-approach, ek satellite-index, is sawaal ko poochne ka ek-akela reasonable tareeka nahi hai, aur kam-se-kam ek dosra reasonable tareeka, ek alag built-up-classifier, usse sehmat nahi hai — aur ye disagreement kisi bhi check se abhi tak explain-away nahi hua.

Is scheme ki scale aur strategic-framing dekhte hue, ise us independent impact-assessment ke bina continue nahi karna chahiye jiska iska apna parliamentary-record confirm karta hai ki kabhi hua hi nahi — aur ye study khud ab ek demonstration hai ki wo assessment kyun care maangta hai, ek se zyada measurement-convention, aur checks jo theek se re-verify hoti hain, sirf pehli pass par bharosa nahi. Ek incomplete extraction, ek contaminated control-group, aur ek snow-contaminated cloud-mask — har ek ne independently ek result diya jo tab tak

robust dikha jab tak underlying problem mila aur fix hua. Ek alag proxy ne independently ek result diya jo is study ke primary finding se disagree karta hai aur, baaki teen ke ulta, abhi tak explain-away nahi hua hai. Koi bhi future assessment, official ho ya independent, usi standard par hold ki jaani chahiye jo ye study khud par lagati rahi: complete, decontaminated data, ek consistent timeline par pull ki gayi, ek control-group ke against check ki gayi, ek buffer-radius sweep, ek multi-year comparison, ek se zyada satellite-proxy, aur ek se zyada cloud-masking-convention — har ek theek se re-verify hoti hui jab bhi external-review jaisa koi check doubt karne ki wajah de. Ye kisi bhi ek window, ek extraction, ek proxy ke result par bharosa karne se behtar hai, jo pehle mil jaaye.

References — Ek Chhota Overview

Ye paper apni buniyad kayi tarah ke sources par rakhta hai: theory (Buzan, Waever, aur de Wilde, 1998, securitization theory ke liye), remote-sensing methodology (Zha, Gao, aur Ni, 2003 NDBI ke liye; Elvidge et al., 2017 VIIRS ke liye), statistics (Wilcoxon, 1945; Spearman, 1904; Dutilleul, 1993, Section 6.11 ki spatial-correction ke liye; Angrist aur Pischke, 2009, Section 3.7 ke DiD-specification ke liye), aur paanch primary government-sources — Parliament ke Rajya Sabha aur Lok Sabha Q&A; documents jo PDF 20 mein Section 2.2 ke saath detail mein cover hue. Reference-list khud bhi is paper ki imaandaar-history ka hissa dikhati hai: Question No. 3251 ki entry mein seedha wahi 2026-09-18 citation-audit-correction likhi hai jo Section 2.2 mein describe hui — purani, galat citation (Question No. 401, Wayanad landslide) ko bhi disclose karte hue, sirf sahi citation dikhane ke bajaye. Ye poore paper ke ek chhote se hisse mein wahi pattern dikhata hai jo ye teaching-series shuru se seekha rahi hai: galtiyan chhupayi nahi jaati, record ki jaati hain.

RECAP — PDF 21 Mein Kya Seekha: (1) Section 4 Results — 4.2 ka poora H1 null (aur ek naya detail: full-year window mein 75.7% villages decrease dikhate hain), 4.3 night-lights stable null, 4.4 budget-sign-flip, 4.5 H3 border-proximity ka naya summer-result, 4.6 ki poori control-group DiD kahani (753 se 721 villages tak), 4.7 ka non-monotonic decline-phir-recovery pattern, 4.8 buffer-radius null, 4.9 Holm-Bonferroni (2 survivors, dono naye), 4.10 SAR-agrees/Dynamic-World-disagrees triangulation. (2) Section 5 Discussion — do reversals ki meta-seekh, aur ek 2026-09-18 internal-contradiction correction. (3) Section 6 Limitations, 6.1 se 6.12 — village-gaps, border-geometry-nuance, geocoding district-selection-bias, ground-truth reframing, baseline-imbalance ka flip-flop, threats-table, spatial-autocorrelation-correction, aur cloud-mask-bug ki poori kahani. (4) Section 7 Future Work, 7.1 se 7.7 — zyaadatar ab 'completed' hai: SAR, building-footprint, ground-truth (Walong/Kaho/Musai ka asli sourcing), pre-treatment-placebo, H3 stress-testing (linearity vs temporal-stability), aur statistical-power/MDE. (5) Section 8 Conclusion — teen items jo ek jaisi dikhti hain lekin nahi hain: do resolved-artifacts (Sikkim missing-data, control-contamination), ek asli-unresolved-tension (SAR-vs-Dynamic-World), aur ek resolved-bug (QA60-vs-SCL).

Aage, PDF 22 mein — is 22-PDF series ka aakhri PDF — poori BORDER_OPTICS research ko ek connected, current, corrected big-picture kahani ki tarah synthesize kiya jaayega, pitch-ready confidence ke saath.